



Technology and writing: Review of research

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ABSTRACT

This review examined 29 empirical studies published in peer-reviewed journals from 2002 to 2017 that investigated the use of computers and information and communication technologies during writing instruction and related writing activities. Qualitative content analysis was employed to investigate how technology was used in the writing program, the impact of technology-mediated instruction on students' writing skills, and the barriers teachers faced in integrating technology into the writing curriculum. Results of the review demonstrated that computers and a range of digital technologies, applications, and web-based learning environments have been used to teach writing in K-12 educational settings. Technology-mediated writing instruction yielded improvements in students' composing processes and writing skills as well as their knowledge and use of new literacies. Students designed, produced, and presented a variety of multimodal and digital texts that represented their knowledge and understandings of literary material and contemporary social justice issues. The use of technology motivated student engagement and participation in writing assignments and increased social interaction and peer collaboration. Teachers faced a number of challenges in their efforts to integrate technology into the writing curriculum. Several recommendations are offered based on the results of this review. Relevant, high-quality teacher professional development on pedagogical uses of technology are urgently needed in order to promote technology-mediated writing instruction and build students' 21st century literacy skills. Institutional support is needed to ensure the availability of computers and appropriate applications in every classroom. Future research needs to identify how teachers can use multimedia and digital tools to improve writing instruction in the digital age, meet the objectives of the writing curriculum, and engage all students in the composing process. Continued exploration of how digital writing environments and online writing communities can support students' competence as writers also is needed. Substantial investigative work is needed in the early childhood program.

1. Introduction

Learning to write is fundamental to becoming literate, and proficiency with writing is crucial to academic achievement, employment, and promotion in the workplace (Graham & Perin, 2007). For most students, learning to write and becoming a competent writer requires deliberate instruction (Raban & Scull, 2013), which typically occurs in school settings. In light of the rapid development and accessibility of information and communication technologies (ICTs), the definition of “writing” and the nature of writing instruction need to be reconsidered. While many students are interacting in social cyber networks and using multimedia and digital

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tools to communicate in their everyday lives—particularly those students from higher socioeconomic classes (Coiro, Knobel, Lankshear, & Leu, 2008)—their school writing experiences are often quite different. In some classrooms, students are still composing with paper and pen and interacting solely with print-based texts (Alvermann, 2008; Black, 2009; Moje, Overby, Tysvaer, & Morris, 2008). In other classrooms, digital composition is increasingly becoming an integral part of the writing program (Mills, 2014).

Teacher educators and educational researchers are in a position to influence pedagogy, practice, and research; thus, it is important to be (a) well-informed about how digital technology has been used in the writing program, (b) knowledgeable about whether and how its use has supported the development of students' writing skills, and (c) aware of the barriers teachers face in implementing technology-mediated writing instruction. To that end, we synthesized research studies from the past 15 years that examined the use of computers and ICTs during writing instruction. Our review is timely, given the changing nature of literacy in our technology-saturated society and the need to develop students' 21st-century writing skills.

2. Research questions

Three research questions guided our work:

- (1) How is technology used to mediate writing instruction and related writing assignments among K-12 learners?
- (2) How does technology-mediated writing instruction and related writing assignments support the development of students' writing skills?
- (3) What barriers do teachers face in moving toward technology-mediated writing instruction?

3. Method

To locate studies for consideration in this review, we first used several electronic databases focusing on technology, including ACM Digital Library, Computer & Applied Sciences Complete, GALE Computer Database, and IEEE Xplore. Because we were primarily interested in the use of technology to mediate writing, we also searched ERIC, Education Full Text, Education Research Complete, and EditLib Digital Library. We employed multiple configurations of various content search terms, *technology, information and communication technology, ICT, writing, writing instruction, and composition* and population search terms, *early childhood, elementary, middle school, junior high, and high school*. We limited our search to articles published in English. Using a snowball search strategy, we examined the reference list of each study we identified in order to locate additional relevant studies.

3.1. Inclusion criteria

In a previous review of research, Andrews et al. (2007) examined the effectiveness of ICTs on 5-to-16-year-old students' writing. They limited their investigation to controlled trials published between 1998 and 2003 with the goal of establishing evidence-based practice. The researchers identified only nine studies that met their inclusion criteria. They concluded that their narrative analysis offered “no consistent view” (p. 333) due to the heterogeneity of the nine studies and variability in definitions of the terms *ICT* and *writing* across the data set. The researchers did, however, provide brief summaries of each study, including major findings, which may be informative to readers.

We established the following inclusion criteria for each study: (a) published in a peer-reviewed journal between 2002 and 2017, (b) examined the use of technology to teach writing and/or investigated K-12 students' use of technology to compose texts, and (c) were data-driven empirical studies (i.e., specified data sources). We intentionally chose a 15-year timeframe (i.e., 2002–2017) in order to conduct a comprehensive yet relatively recent review of the literature. Only studies that examined teachers' and students' use of technology for writing were included. For example, studies that examined the potential of a specific technology (e.g., stylus) but did not investigate its use in the context of teaching and learning to compose text were excluded. Studies that examined technology and literacy were excluded, unless the study examined both reading and writing. We applied these criteria to the abstract of each article we identified and consulted the full text as needed to determine inclusion or exclusion. A total of 29 studies met these criteria and were included in this review.

3.2. Analysis procedures

We analyzed the data inductively using conventional content analysis (Hsieh & Shannon, 2005) in order to answer our research questions and potentially uncover additional insights about the use of technology to teach writing. Conventional content analysis is highly systematic; first, we read each article in its entirety to gain an overall conceptualization of the research investigation. Second, we identified each study's purpose and instructional pedagogies in order to answer our first research question on how technology was used to teach writing and mediate students' writing assignments. Third, we located each study's major findings to address our second research question on how technology-mediated instructional activities supported the development of students' writing skills. Fourth, to answer our third research question, we searched the entire data corpus looking for reported barriers or challenges teachers faced in integrating technology into writing instruction.

Finally, we searched the entire data set for words or phrases that were repeated across multiple studies; such repetition might signal importance and point us to additional insights about technology-mediated writing instruction. During this fifth and final wave of analysis, three themes emerged with regard to the use of technology in the writing program: *motivation, collaboration, and reluctant*

Table 1

Number of studies addressing RQ 1, RQ 3, and Additional Findings.

RQ 1	Subtheme	# of Studies				
		K	Primary	Elementary	Middle/junior high	High school
Classroom-based technology-mediated writing		3	3	2	1	4
Online Learning Environments	Online Tutoring			1		1
	Web-based writing			2		
	Hypermedia authoring				1	
Additional Findings		# of Studies				
Technology and Motivation		13				
Technology and Collaboration		18				
Technology and Reluctant Writers		6				
RQ 3	Subtheme	# of Studies				
Barriers to Using Technology for Writing Instruction	Teacher beliefs	5				
	Need for Professional Development	7				
	Access to ICTs	6				

writers. We subsequently explored the significance of those themes across the data corpus and uncovered several additional findings.

In the sections below, we report the results of our analysis. First, we share the findings to our research questions. Then, we discuss the three additional themes that emerged. As did [Andrews et al. \(2007\)](#) in their review, we provide a brief description of the studies in order to illuminate the specific technologies, instructional pedagogies, and research methods that were employed. Then, we provide a synthesis of the findings across studies. In the final section of the paper, we discuss the significance of our results and offer implications for pedagogical practice and future research.

4. Results

4.1. Research questions 1 and 2

For the sake of clarity, we distinguished between investigations that were conducted face-to-face in K-12 classrooms and investigations that were conducted within an online learning environment. Thirteen studies included in this review investigated classroom-based technology-mediated writing instruction (see [Table 1](#)). Five studies examined writing instruction in an online learning environment. We briefly discuss each study in the sections below, highlighting major findings.

4.1.1. Classroom-based technology-mediated writing instruction

All 13 studies that examined face-to-face technology-mediated writing instruction were conducted in K-12 classrooms. Three of the studies were conducted in kindergarten, three in primary school, two in elementary school, one in middle school, and four in high school.

4.1.1.1. Technology and emergent literacy in kindergarten. In three studies, researchers examined the ways in which technology mediated kindergarten children's emergent reading and writing skills. Using experimental designs, [Voogt and McKenney \(2007\)](#) and [McKenney and Voogt \(2009\)](#) investigated how the computer software application PictoPal supported emergent literacy skills in 4- and 5-year-old children. PictoPal is designed to help children understand the nature and function of written language and the relationship between spoken language and printed text. PictoPal engages children in closed-question (i.e., reading pictogram/rebus sentences, sight-word recognition) and open-question activities (i.e., writing lists and personal letters). During open-question activities, children often created documents to be used for authentic purposes in the kindergarten classroom (e.g., recipes for the kitchen or menus for the restaurant in the dramatic play center). The experimental group (N = 21) worked with PictoPal for four sessions over a 5-week period; the control group (N = 19) participated in class-as-usual literacy instruction. Results indicated that after some instruction from an adult, kindergarteners in the experimental group were able to use the mouse, adjust to the computer voice, and work independently with the computer program. However, the PictoPal intervention did not improve students' emergent literacy skills. In a second study, the researchers increased the number of PictoPal sessions to eight, included only open-question activities, and specifically designed those activities to connect directly to the current classroom theme. Results demonstrated a statistically significant learning effect for the experimental group (N = 7) on a measure of early literacy as compared to the control group (N = 7). Both teachers and students indicated that the activities which were related to the classroom theme were more motivating, challenging, and meaningful. Findings also revealed that while the children were keen on using the computer, they were particularly eager when a parent volunteer was available to assist; the interaction between the adult and child encouraged dialogue and reflection on the feedback tools in the PictoPal software.

In a subsequent study, [McKenney and Voogt \(2009\)](#) documented teachers' use of students' PictoPal products in kindergarten

activities and examined the impact of the pedagogical practice on the children's emergent literacy skills. The researchers found that over time teachers were better able to connect on-computer activities and children's PictoPal products to the early literacy curriculum. Results also demonstrated that the experimental group ($N = 39$) achieved significantly higher emergent literacy gains than did the control group ($N = 40$). The researchers posited that using students' PictoPal products in dramatic play centers and for other authentic purposes in the classroom was essential to supporting early literacy learning.

More recently, [Beam and Williams \(2015\)](#) investigated one kindergarten teacher's use of ICTs to mediate early writing instruction. In a qualitative case study, the researchers documented the nature and content of the teacher's technology-mediated instruction and children's appropriation of that instruction to support their independent writing. Results of the study revealed that the teacher used paper and pen and a range of ICTs (e.g., videos, smartboard, document camera, digital pen and writing pad) to demonstrate how a writer composes texts. He explicated a process approach to writing (i.e., planning, drafting, receiving feedback, revising, editing, publishing), taught a range of writing strategies, and discussed characteristics of specific genres. Students were attentive and motivated to participate during writing instruction, given their fascination with the technology and the multimodal texts their teacher created. Their engagement in the lessons supported their appropriation of their teacher's instruction. During independent writing, the kindergarteners composed hand-written texts using a process approach, supported by the strategies the teacher had taught. The teacher often used PhotoStory to create multimodal representations of the children's stories. PhotoStory allows users to import photos, record custom narration, and export the final product as a video file. Using his cell phone, the teacher took photos of the children's drawing/writing and then recorded the children "reading" their compositions. He imported these into PhotoStory to create videos that were shared at the end of the writing lesson. Children were eager to have their writing published via PhotoStory. Sharing their writing via video provided unique opportunities for student-authors to discuss and receive feedback from classmates.

4.1.1.2. Technology and writing in primary school. [Mills \(2011\)](#), [Kervin and Mantei \(2016\)](#), and [Yamac and Ulusoy \(2016\)](#) investigated young children's use of technology to plan, produce, and share digital texts in the primary-grade writing program. All three studies were conducted in third grade. [Mills \(2011\)](#) was most interested in how 8-year-old children made connections between different modes of communication (i.e., written words, images, music, video) as they created digital and multimodal texts in response to children's literature. She called this process "transmediation" because the modes do not have exact one-to-one correspondences. Teachers in the project received on-going professional development in planning media-based writing lessons and introduced students to a range of media software, blogs, podcasts, micro-documentaries, web profiles, digital stories, and online comics. Mills used observation, audio-recorded focus groups, and analysis of students' multimodal texts to investigate the processes of transmediation. Findings revealed that as the children moved among modes to create a range of multimodal texts, the opportunities for generative thinking were increased. For example, creating a digital comic based on a children's novel engaged students in transmediation, because the modes (i.e., drawing, written text, digital images) do not have precisely parallel meanings, and so children had to problem solve in order to express their understandings, especially in light of the text-creation affordances of the technology. Mills argued that transmediation is central to digital text production, and providing students opportunities to produce digital texts in response to children's literature will allow them to transform their knowledge of literary material, rather than simply reproduce it.

[Kervin and Mantei \(2016\)](#) examined the ways in which one 3rd-grade student used a range of iPad applications and a process approach to writing to create a "work-flow" (p. 135) that supported his completion of an inquiry project. The student used the *Popplet* application to plan his text, the interactive search application *instaGrok* to gather information on his topic, *Google images* for photos, and *Tellagami* to create his own avatar. He also created an animated video. He drafted his text in his writing book and conferenced with his teacher to revise, redraft, and edit. Then he typed it in *Book Creator*. His teacher used *Reflector* to share his project with classmates allowing him to receive peer feedback. The student then published his multimodal text in *iBooks*. The researchers argued that the affordances of the technology, and the supportive structure of a process approach to writing, empowered the young child to create multimodal texts that effectively demonstrated his knowledge and understandings of his inquiry topic.

[Yamac and Ulusoy \(2016\)](#) examined the effectiveness of digital storytelling in improving third graders' writing skills. Storytelling has been used for decades to support students' literacy development by activating their reading, writing, listening, and speaking skills. Digital storytelling actuates those same skills, as well as new literacies skills, as children integrate images, graphics, music, audio, and video into a coherent narrative with a theme, plot, and viewpoint, just as in traditional stories. Teachers in the project introduced digital stories and taught children how to create them. Across the 8-week intervention, 26 students created digital stories in groups, then in pairs, and lastly individually. They used a process approach to writing, storyboarding, and multimedia from the Internet to draft their compositions. Then, they used PhotoStory to publish their digital stories, which they shared with classmates and online. Researchers used pretest–posttest comparisons to reveal statistically significant differences between students' baseline and post-intervention stories on measures of ideas, organization, word choice, sentence fluency, conventions, story elements, and number of words. Creating digital storyboards helped children to better understand sequence of action as students divided their stories into parts, drew or imported pictures for each part, and then wrote narration for each image. Creating spoken narration and then listening to the audio allowed students to check the flow and accuracy of events and revise as needed to further develop plot structure. Editing also was facilitated by the recursive phases of storyboard production. Use of the technology made the recursive nature of the writing process transparent. Moreover, students' new literacies skills improved across the intervention. Observation and interview data demonstrated that students were motivated to use the digital storytelling tools. Students who had been reluctant writers showed enthusiasm to participate in digital writing processes and gained self-confidence through collaborative work. Stories created collaboratively were stronger in quality than stories created individually.

4.1.1.3. Technology and writing in elementary school. In two studies in the elementary grades, researchers investigated the impact of

ICTs on students' expository or narrative writing skills. Doan and Bloomfield (2014) used an experimental design to examine how search time on the Internet influenced students' expository essay scores. Forty-nine 4th and 5th-grade students were randomly assigned to three groups and given the same writing prompt. Group I (control group) received the writing prompt and 90 min to compose their essay. Group II received 30 min of search time on the Internet to research their topic and 60 min to write. Group III received three 45-min training sessions prior to writing day that addressed Internet domain names, using key words during searches, and evaluating a website for authenticity; then, on writing day, they received 30 min of search time on the Internet and 60 min to write. Groups II and III used notebook computers. Student essays were evaluated for ideas, (i.e., main and supporting ideas, arguments, organization), written expression (i.e., word choice, tone, voice, sentence structure), and mechanics (i.e., spelling, capitalization, punctuation). Results of an ANOVA analysis demonstrated that Groups II and III, who used the Internet during prewriting, demonstrated higher mean total essay scores than the control group ($p = 0.053$), almost reaching statistical significance. Group III, who received training on how to use the Internet, achieved higher essay scores than did Group II; effect sizes ranged from .22 (small) to 0.83 (large). The researchers then combined Groups II and III, who used the Internet during prewriting, and compared it to the control group, and found statistically significant differences in total essay score and mechanics. Overall, students in both experimental groups scored higher than students in the control group, suggesting that access to the Internet for research purposes can improve students' expository writing. The researchers hypothesized that use of the Internet lessened students' cognitive load by making it easier for students to gather information and refer to it repeatedly as needed during the composing process. They suggested that use of the Internet bolstered students' confidence and self-efficacy and thus their writing scores.

Sessions, Kang, and Womack (2016) examined the influence of iPad applications on 5th-grade students' narrative writing. One group of 15 students used paper and pencil to compose while the other group of 15 used iPad applications (e.g., Toontastic, Story Builder, Dragon Dictation). The teacher presented lessons aimed at improving students' narrative writing with respect to visualizing, sequencing, and including details. Researchers analyzed the students' writing at the end of the 9-week intervention and found that both groups showed improvement on all three targeted skills when compared to baseline. Both groups made similar progress with regard to incorporating details, but the iPad group demonstrated stronger visualization skills and wrote stories that were more cohesive and sequential. Using the iPad, students were able to create picture sequences on a single screen, giving them a coherent view of the story, which they used to guide narrative composition. Students using paper and pencil wrote stories as ideas came to mind, rather than sequencing events in logical order. In exit interviews, students reported, and the teacher observed, that the iPad applications motivated students to actively engage in writing tasks, which increased their self-confidence and supported improvement in their writing skills. Use of the iPad applications also increased social interaction as students were drawn toward collaboration with their peers.

4.1.1.4. Technology and writing in middle school. Hitchcock, Rao, Chang, and Yuen (2016) investigated the effectiveness of a technology-integrated writing intervention, TeenACE for Science, on middle school students' expository writing skills. Teachers taught the major components of a science report and explicated a process approach to writing. Across the 12-week intervention, students composed science reports individually or collaboratively. Students used the multimodal features of PowerPoint as a tool for organizing their writing. For example, they used PowerPoint slides to record the outcomes of an experiment or take notes on information they had gathered about a science topic. They imported pictures onto slides and created headings. Then, they used those images as prompts for writing sentences and paragraphs. When their first draft was complete, they recorded themselves reading what they had written on the slides. Then, they listened to the audio-recording and redrafted, revised, and edited their report as needed. Students then cut and pasted their compositions into a word processing program to create their final report. Students and teachers evaluated the reports using a writing rubric, and students shared their presentations in a science fair. Results of a pretest-posttest comparison showed significantly higher scores on standardized assessments of writing after the intervention. Results on curriculum-based assessments revealed improvement in students' ability to outline, organize, select appropriate words, and, evaluate, revise, and edit their writing. During interviews, students stated that they enjoyed using the computer and appreciated opportunities to be creative. Use of the technology increased their motivation, engagement, and self-confidence; students began to see themselves as competent writers. Teachers stated that students benefited from the individualized instruction they were able to provide through the process approach to writing and from collaboration with peers.

4.1.1.5. Technology and writing in high school. Two of the four studies conducted at the high school level examined students' use of technology to represent their understandings of pop-culture literature and contemporary social justice issues. The other two studies investigated students' use of technology to improve their argumentative writing skills.

Lawrence, McNeal, and Yildiz (2009) used multimedia and pop-culture texts to support reading, writing, and new literacies skills in 12 urban high school students who attended a three-week, university campus-based summer program. University faculty introduced a range of digital literacy skills and taught specific strategies for reading, comprehending, and writing visual texts (i.e., comics, graphic novels) and engaging in the research process. Students then practiced those strategies and skills on the computer in groups and independently. During the 3-week program, they created a comic strip about themselves, a response to a graphic novel, a critique of a graphic novel (which they uploaded to a bookstore website), and a group research project in which they gathered information from the Internet and created an informational comic strip on a contemporary social justice issue (e.g., modern-day slavery). Students used PowerPoint, word processing, Comic Life software, images and information from the Internet, and photos taken with a digital camera. An analysis of the students' products revealed that students had increased their knowledge of and ability to use technology, developed their ability to critically analyze and synthesize informational text which contributed to improvements in their writing, and expand their understanding of how to communicate using visuals and texts. The group research projects

demonstrated students' knowledge of the social justice issue they had explored. Students were engaged in the computer-based tasks and motivated to use the technology.

Jocius (2013) investigated 12th grade students' use of digital media to demonstrate their understandings of literature in an advanced-placement English course. The students chose to work individually or collaboratively to create multimodal representations of the novel *The Kite Runner* (Hosseini, 2003), which depicts the turbulence in Afghanistan in the 1970s. Students also chose the format for their representation (e.g., timeline connected to real world events, retelling, literary analysis) and the digital composition tool they would use, either PowerPoint, Keynote, iMovie, or MovieMaker. Findings of a qualitative analysis indicated that students' multimodal representations demonstrated their analytic and interpretative skills and their understanding of the novel. The type of digital tool students used had a direct impact on the content students presented and the modes that were purposefully featured. For example, all students working with PowerPoint included quotes from the novel and written explanations of literary devices (e.g., metaphor, flashback), while all students composing with digital video labeled literary devices but used music in place of written explanation. In addition, the modes students employed influenced the tone and mood (e.g., humorous, dramatic) of their presentations. Results also indicated that student choice in working independently or collaboratively, in format of the representation, and in digital tool afforded students' more ownership of their final product. Moreover, 94% of students stated that they would prefer to represent their literary understandings through multimodal presentations rather than traditional essays. Jocius argued that students gained a deeper understanding of the novel through their research into events in the novel and subsequent selection of images and videos to represent those events.

In Turner and Katic's (2009) investigation, sophomore students used a computer and a word processing program and had access to the Internet as they composed argumentative essays in a college-prep English class. The researchers examined the process of students' writing as well as students' appropriation of new literacies skills. Findings revealed that students' writing was "distinctly non-linear" (p. 264). Use of the technology allowed students to engage in the recursive nature of writing with ease, and students' non-linear approach became automatic as they internalized the affordances of the ICTs. Turner and Katic argued that students began to think and write fluently in ways that reflected the non-linear framework of the technology.

Howell, Butler, and Reinking (2017) examined the effectiveness of an 11-week intervention that taught high school students to construct multimodal arguments using digital tools with the goal of improving their ability to write conventional arguments. The teacher introduced the digital tools (e.g., iPads, Glogster EDU, websites, Evernote), taught the elements of arguments (i.e., claims, evidence, and warrants), and provided examples of arguments in models of writing. She then helped students to link argumentation to multimodality by analyzing how advertisements conveyed elements of argument in conventional form and in multimodal form (e.g., images, sound effects, music). She also discussed differences between developing conventional written arguments and constructing multimodal arguments. Students then wrote conventional arguments and created multimodal arguments about social issues of personal interest using a recursive, process approach to writing. Researchers compared students' baseline arguments with post-intervention arguments and found no statistically significant differences; there was little evidence to suggest that constructing multimodal arguments had transferred to writing conventional arguments. However, students' multimodal products suggested that they had extended their understanding of and ability to construct an argument. Students' inexperience with extended writing, particularly argumentation, and the digital tools may have inhibited the intervention's effectiveness. Observation data indicated that a process approach to writing allowed the teacher to individualize instruction and provided students opportunities to experiment with multimodal writing in general, and formulating multimodal arguments in particular, at their own pace. In interviews, students stated that they were motivated to participate and preferred designing multimodal arguments over conventional arguments. The teacher observed students' motivation and increased engagement and participation; she noted that students who had been reluctant writers were writing markedly more.

4.1.1.6. RQ 1 synthesis: classroom-based technology use. A range of ICTs have been used for a variety of writing-related purposes in K-12 classrooms (see Table 2). For example, in kindergarten, teachers used ICTs to teach emergent literacy and early writing skills (McKenney & Voogt, 2009; Voogt & McKenzie, 2007) and to transform children's handwritten stories into digital texts (Beam & Williams, 2015). In the primary and elementary school grades, students used ICTs to compose narratives (Sessions et al., 2016; Yamac & Ulusoy, 2016). Students in primary and middle-school grades used ICTs to compose informational and expository texts (Hitchcock et al., 2016; Kervin & Mantei, 2016). Students in high school used ICTs to compose argumentative essays (Howell et al., 2017; Turner & Katic, 2009). In the elementary grades as well as in high school, students conducted research on the Internet to support informational and expository writing (Doan & Bloomfield, 2014; Lawrence et al., 2009; Turner & Katic, 2009). Students in the elementary, middle, and high school grades used ICTs to organize their research and to create drafts of their reports and essays (Hitchcock et al., 2016; Lawrence et al., 2009; Sessions et al., 2016). Students in the primary, middle, and high school grades used a range of ICTs to create an array of multimodal products to communicate their ideas or demonstrate their knowledge and understandings of literary material and inquiry/research projects (Howell et al., 2017; Jocius, 2013; Kervin & Mantei, 2016; Lawrence et al., 2009; Mills, 2011; Turner & Katic, 2009). In kindergarten and the primary grades, students used ICTs to share their writing and receive feedback from peers and wider audiences (Authors; Kervin & Mantei, 2016; Yamac & Ulusoy, 2016). Students in the primary, middle, and high school grades used word processing to type their final compositions in preparation for publication (Hitchcock et al., 2016; Kervin & Mantei, 2016; Lawrence et al., 2009; Turner & Katic, 2009).

4.1.1.7. RQ 2 synthesis: impact of technology. The use of technology in the K-12 classroom-based writing program supported improvements in students' writing. For example, when kindergarten teachers directly related computer activities to the early literacy curriculum, students made statistically significant improvements in emergent literacy and early writing skills (McKenney & Voogt,

Table 2
Uses of ICTs in K-12 classrooms.

Goal	ICTs	Grade levels	Studies
Teach emergent literacy and early writing skills	Photostory, digital writing pad and pen, interactive whiteboard, document camera PictoPal	Kindergarten	Beam & Williams, 2015; McKenney & Voogt, 2009 Voogt & McKenney, 2007
Compose narratives	iPads and writing applications Digital storytelling using Photostory 3 software	Kindergarten, Primary, Elementary	Sessions et al., 2016 Yamac & Ulusoy, 2016
Compose informational/expository texts	Software, TeenACE for Science (TAS) iPads, Google Image search, and applications (Popplet, Book Creator, instaGrok, Tellagami, Reflector, PuppetPals, iMovie, Quicktime)	Primary, Middle school	Hitchcock et al., 2016 Kervin & Mantei, 2016
Compose argumentative texts	iPads, Chromebooks, laptop computers, Google Applications, and Glogster websites Computers with Internet access	High school	Howell et al., 2017 Turner & Katic, 2009
Conduct research and gather information from the Internet	Computers with Internet access	Elementary, High school	Doan & Bloomfield, 2014 Lawrence et al., 2009 Turner & Katic, 2009
Organize research; create drafts of reports/essays	Software, TeenACE for Science (TAS) Computers with Internet access iPads and writing applications	Elementary, Middle school, High school	Hitchcock et al., 2016 Lawrence et al., 2009 Sessions et al., 2016
Create an array of multimodal products to communicate ideas or demonstrate knowledge and understandings of literature and inquiry/research projects	iPads, Chromebooks, laptop computers, Google Applications, and Glogster websites Computers with the ability to create multimodal projects (i.e. iMovie, MovieMaker, PowerPoint, Keynote) iPads, Google Image search, and applications (Popplet, Book Creator, instaGrok, Tellagami, Reflector, PuppetPals, iMovie, Quicktime) Computers with Internet access Teachers introduced students to blog pages, podcasts, micro-documentaries, web profiles, digital stories, online comics, and a suite of Apple media software. Students created various digital artifacts (i.e. digital drawings, storyboards, scripts, digital movies, comics)	Primary, Middle school, High school	Howell et al., 2017 Jocius, 2013 Kervin & Mantei, 2016 Lawrence et al., 2009 Mills, 2011 Turner & Katic, 2009
Share writing and receive feedback from peers and wider audiences	Digital writing pad and pen, interactive whiteboard, document camera iPads, Google Image search, and applications (Popplet, Book Creator, instaGrok, Tellagami, Reflector, PuppetPals, iMovie, Quicktime) Digital storytelling using Photostory 3 software	Kindergarten, Primary	Beam & Williams, 2015; Kervin & Mantei, 2016 Yamac & Ulusoy, 2016
Type final compositions for publication	Computers and word processing	Primary, Middle-school, High-school	Hitchcock et al., 2016 Kervin & Mantei, 2016 Lawrence et al., 2009 Turner & Katic, 2009

2009). Primary-grade students demonstrated statistically significant improvements in narrative composition skills when they created digital stories (Yamac & Ulusoy, 2016). High school students' informational writing improved and elementary-grade students made statistically-significant improvements in expository composition skills when they used the Internet to research their topics (Doan & Bloomfield, 2014; Lawrence et al., 2009). Elementary school students wrote more cohesive narratives when composing with iPad applications vs. paper-and-pencil (Sessions et al., 2016). Middle-school students demonstrated significantly higher expository writing skills when they used the multimodal features of PowerPoint to organize their research prior to writing science reports (Hitchcock et al., 2016). High school students' informational writing improved when they used a range of ICTs to create comic strips about a contemporary social justice issue (Lawrence et al., 2009).

Moreover, technology-mediated writing activities also benefitted other aspects of students' learning. For example, to create multimodal texts, primary-grade children had to engage in *problem-solving* and *generative thinking skills* (Kervin & Mantei, 2016; Mills, 2011). The use of digital storyboards and iPad applications to compose narratives supported primary- and elementary-grade students' *sequencing skills* (Sessions et al., 2016; Yamac & Ulusoy, 2016). High school students' *analysis* and *synthesis skills* improved when they researched information on the Internet and used ICTs to create informational comic strips (Lawrence et al., 2009). High school students' use of *analytic* and *interpretative skills* resulted in improved *comprehension* when they researched historical events on the Internet while creating multimodal representations of historical-fiction novels (Jocius, 2013). High school students extended their understanding of *argumentation* when creating multimodal arguments (Howell et al., 2017). In addition, the affordances of the ICTs mediated planning, drafting, revising, and editing, which led to greater *fluency* in the recursive phases of the writing process (e.g., Turner & Katic, 2009; Yamac & Ulusoy, 2016).

4.1.2. Online learning environments

Five studies included in this review investigated the use of online learning environments to support students' writing development

(see Table 1). All five studies were published between 2004 and 2010; we found no studies published after 2010 that examined online learning environments. Two of the five studies explored the use of online tutoring, one with elementary school children and the other with high school students. Two studies examined the impact of web-based writing communities on elementary school children's writing. One study investigated the effect of a hypermedia authoring system on junior-high students' production of web pages.

4.1.2.1. Online tutoring. Holdich, Chung, and Holdich (2004) examined a web-based computer tutor called HARRY that was designed to help elementary school children improve their narrative writing. As the 8- and 9-year-old children composed, HARRY targeted specific weaknesses in grammar and style and provided suggestions for revision. For example, HARRY suggested punctuation for run-on sentences and offered synonyms for repetitive vocabulary use. Each of four case study children wrote a control narrative and a HARRY-assisted narrative on the same theme over a 5-week period. Results of the analysis indicated that the HARRY-assisted narratives were stronger than the control narratives; they included more varied vocabulary, a lower percentage of common words, more sophisticated sentence constructions, and more appropriately-used punctuation. However, performance in some aspects of narrative composition did not improve or deteriorated with HARRY's assistance, most likely due to children paying more attention to certain aspects of their compositions than to others.

Merrill and Rodriguez (2005) examined the use of an online literacy tutoring program designed to help high school students learn how to improve their writing. In the *Digital Divide Project*, 11th-grade students were paired with university graduate students (predominantly in-service teachers) who served as revisers. Students drafted and uploaded their manuscripts into the online tutoring platform, and revisers provided digital feedback in six forms: inserting a comment within the manuscript, commenting only at the end of the manuscript, inserting comments with the insertion tool and placing comments at the beginning of the manuscript, inserting comments with the insertion tool and placing comments at the end of the manuscript, commenting at the beginning and end of the manuscript with no insertions or embedded comments, and embedding comments in the manuscript using a different font color and size than the student had used. The researchers examined the kinds and placement of revisers' feedback to determine the impact on the amount of words students wrote across drafts. Results demonstrated statistically significant differences for three feedback categories: insertions only, insertions and end comments together, and comments written before and at the end of the manuscript. In addition, students wrote statistically significantly more words when revising first drafts than subsequent drafts. Results also demonstrated that students developed a range of new literacies skills using the online tutoring program. Findings of the study suggest that online tutoring programs can support students' growth as writers and their acquisition of new literacies, the nature of teacher feedback matters, and feedback early in the writing process may be most valuable.

4.1.2.2. Web-based writing environments. Yang, Ko, and Chung (2005) developed and evaluated a web-based writing community designed to improve elementary school children's writing. The web-based program offered students three themes and a two-week time frame for completing stories on a chosen theme. The "story pass-on" theme fostered reading comprehension; a children's literature expert started a story and students could take turns adding paragraphs to continue the story. The "story chameleon" theme fostered creativity by asking children to adapt characters, places, and plots as they re-wrote a well-known story. The "thousand ideas" theme taught expository writing; students were asked to write about problems in daily life and how those problems could be solved. The children's literature expert evaluated each story submitted, provided feedback to help students improve their writing, and selected stories to serve as models of "good" writing. Students had access to a self-assessment tool that addressed mechanics, organization, and content, and they also could invite a peer assessment. The assessment process was designed to help students critically appraise their writing and improve their writing skills. Students also had access to discussion boards which addressed writing topics, appreciation for others' writing abilities, being part of a writing community, and reflection on one's writing. Across a two-year period, 2510 students from 257 elementary schools voluntarily participated in the web-based environment, most of whom were 4th-6th graders. Researchers compared students' early and late story submissions and found significant differences for "story pass on" and "thousand ideas" themes. No significant difference was found for "story chameleon," most likely due to the inherent challenges of creative writing. Results also revealed no significant differences between expert and peer assessments; elementary school students understood the criteria and effectively applied them to assessments of other students' writing. Students' self-assessments of their writing were higher than expert or peer assessments, suggesting that children have difficulty identifying weaknesses in their own writing. Moreover, students conducted approximately 20% more peer-assessments than self-assessments. Students reported that they were motivated to participate in the web-based writing environment and highly interested in the interaction within the writing community. Students who once thought writing was difficult came to view it as "much easier than before" (p. 228).

Gnach, Wiesner, Bertschi-Kaufmann, and Perrin (2007) also investigated the use of a web-based writing environment, *myMoment*, designed to promote reading, writing, and new media skills in elementary school children. Children wrote stories directly online, either collaboratively or alone. They chose their writing topics and determined how long they would write in a given session. When they finished their story, they could share it with others to be read, commented on, or evaluated. Children in 1st through 5th grades participated in the study, and two texts from each pupil were analyzed for narrative features. The children were interviewed about the program's interface and how they handled the technology and about their reading and writing habits. Findings of the study indicated that the web-based interactive writing environment influenced children's writing processes. In particular, the *myMoment* program promoted repeated text revision; children returned to their texts on multiple occasions, sometimes using suggestions from peers to improve their compositions. Children also re-read their completed compositions to check for mistakes or misspellings. The children engaged in reader-oriented writing; for example, they inserted Emoticons or multiple exclamation marks as linguistic signals, and they used the computer's formatting features to differentiate their stories from their peers' stories. Students made frequent

use of the comment function, demonstrating their interest in their peers' stories and in giving and receiving feedback. The children stated that they were highly motivated to read and write in the *myMoment* program.

4.1.2.3. Hypermedia authoring system. In a quasi-experimental study, [Chang, Sung, and Zheng \(2007\)](#) investigated the effects of a hypermedia authoring system on junior-high students' production of web pages. The hypermedia authoring system organized the process of creating a web page in ways that reflected a process approach to writing (i.e., setting goals, planning, organizing structure, building content, reviewing). The researchers hypothesized that a process approach would provide students an organizing frame and result in better web pages. At each step of the web-authoring process, students were provided specific authoring tools within the hypermedia system to help them build their web page documents. For example, at the organizing structure phase, the author tools allowed students to edit, delete, insert, and move; at the building content phase, the tools allowed student to insert images, choose fonts and backgrounds, paint, record, and so forth. The classroom teacher provided specific instruction on how to create a web page using the authoring system. Across five 45-min sessions, students created web pages about the Taipei culture based on information they gathered using the research tools in the authoring system. The experimental group had access to all of the authoring tools in the system; the control group had access to data-gathering and editing tools only. The researchers analyzed the students' web pages for content, layout, structure, and links. Results revealed statistically significant differences in all four categories in favor of the experimental group. The guidance provided by the process approach framework and the authoring tools were particularly effective for structure (15.21, $p < 0.01$).

4.1.2.4. RQ 1 synthesis: technology use in online learning environments. Web-based computer tutoring programs have been used to support elementary school children's narrative composition skills ([Holdich et al., 2004](#)), and online tutoring from adults has been used to support high school students' writing ([Merrill & Rodriguez, 2005](#)). Web-based writing communities have been used to improve both primary and elementary school students' writing ([Gnach et al., 2007](#); [Yang et al., 2005](#)). A hypermedia authoring system has been used to improve junior-high school students' production of web pages ([Chang et al., 2007](#)).

4.1.2.5. RQ 2 synthesis: impact of technology. Elementary school students wrote stronger narratives when using a web-based computer tutor ([Holdich et al., 2004](#)), and they made statistically significant improvements in both narrative and expository writing when participating in an online writing community ([Yang et al., 2005](#)). Both primary and elementary school students increased the amount of editing and revising of their manuscripts when they were writing in a web-based writing community ([Gnach et al., 2007](#)). Junior-high school students demonstrated statistically significant differences in content, layout, structure, and links when creating web-pages using authoring tools in a hypermedia authoring system ([Chang et al., 2007](#)). High school students demonstrated statistically significant improvements in their writing when adult online tutors provided feedback in the form of insertions only, insertions and end comments together, and comments written before and at the end of the manuscript ([Merrill & Rodriguez, 2005](#)). Moreover, the high school students' use of the online learning environment also mediated their acquisition of new literacies ([Merrill & Rodriguez, 2005](#)).

4.2. Research question 3

4.2.1. Barriers to using technology

Answers to Research Question 3, the reported barriers or challenges teachers faced in implementing technology-mediated writing instruction, were found across the data corpus.

We identified three primary barriers: (1) teacher beliefs about technology, (2) the need for relevant professional development, and (3) access to technology for instructional purposes (see [Table 1](#)). Each of these barriers/challenges is discussed below.

4.2.1.1. Teacher beliefs about technology. In five studies included in this review, researchers reported that teachers' beliefs about technology influenced their willingness to use ICTs to mediate writing instruction and related writing activities. This theme is best elucidated by [Peterson and McClay \(2012\)](#) who found that Canadian 4th through 8th grade teachers used technology (e.g., computers, LCD projectors, smart boards) to mediate writing instruction but seldom allowed students to use ICTs to complete writing assignments. Two primary assumptions prevented the teachers from taking full advantage of the affordances of digital technology for writing. First, the teachers believed that using a computer and word processor placed greater cognitive demands on students than using paper and pen to organize and draft a manuscript. Second, the teachers believed that students' spelling development would be hindered by the word processor's spell check. In these classrooms, students wrote multiple drafts using paper and pen; they checked for correct spellings, received feedback from teachers and peers, and revised and edited. Only when the manuscript was fine-tuned did they type it on the computer for a polished, final product. One 8th grade teacher noted that his students preferred to compose on computers and thought nothing of creating "a full-blown multimedia presentation" (p. 142); yet, in their writing classes, these students composed by hand. The teachers assumed that students had to master conventional writing processes before they could learn to use technology for writing. Interestingly, the teachers allowed students with special needs and learning disabilities to compose on computers. Apparently, the teachers did not recognize this contradiction in their beliefs and practices.

Other studies have reported similar results. [Gremiger \(2003\)](#) found that high school teachers in Argentina also believed that word processing slows down the writing process due to the cognitive strain of keyboarding on composing processes; consequently, they did not permit word processing in school. Several studies reported that teachers believed digital writing practices were too time-consuming to develop and kept them from addressing the objectives of the curriculum ([Ihmeideh, 2009](#); [Mills & Exley, 2014](#); [Vrasidas,](#)

2015). Vrasidas (2015) found that while 95% of primary school teachers in Cyprus stated that they believed in the importance of integrating ICTs into their instruction, very few teachers did so; instead, they used technology to plan their lessons and prepare teaching materials. They did not develop technology-mediated writing activities, nor did they provide opportunities for students to use technology to support learning; instead, students used technology to play educational games. These findings corroborate those of previous research in which literacy teachers rated the importance of integrating digital tools into their instruction well above their reported level of use (Hutchison & Reinking, 2011). Indeed, the evidence suggests that even when technology is accessible, teachers have found reasons not to use it (Aldunate & Nussbaum, 2013).

4.2.1.2. The need for relevant professional development. Seven studies included in this review illuminated the role of professional development in supporting teachers' movement toward technology-mediated writing instruction. In five of those studies, researchers reported that teachers' attitudes toward and willingness to use ICTs was influenced by inadequate professional development on how best to integrate technology into instructional activities (e.g., Ihmeideh, 2009; Peterson & McClay, 2012). Teachers in Gremiger's (2003) study, for example, reported that previous professional development had been technological (i.e., focused on hardware) rather than pedagogical (i.e., focused on instruction) in nature. Teachers in Vrasida's (2015) investigation expressed frustration over the absence of their voices in decision-making regarding the instructional content of professional development training they were offered.

In a qualitative investigation of a two-week summer professional development program, Sanchez (2011) uncovered some of the challenges teachers faced as they explored technology integration for writing instruction. The summer intensive was designed to replace the one- and two-day intermittent professional development workshops held throughout the school year. Findings of the study indicated that the two-week intensive was fraught with problems. Initially, for example, participants had difficulty with online login registration, accessing the wireless network, and signing up for an online Google discussion. Those technology issues (which could have been resolved prior to the start of the program) took precedence over a focus on using technology for instructional purposes, which fueled teacher frustration. Moreover, some teachers grew frustrated with lengthy demonstrations of particular technologies when they had hoped for creative engagements in technology-mediated instructional projects. In addition, the two-week timeframe made it challenging for teachers to complete their digital storytelling and video production assignments using PhotoStory and MovieMaker.

More positive results were reported by Edwards-Groves (2011) who investigated an 18-month professional development initiative designed to help elementary school teachers integrate ICTs into the writing curriculum. The 17 teacher-participants taught students in kindergarten through 6th grade. Professional development sessions focused on software demonstrations, guided practice in using ICTs, understanding elements of design, model lessons, and peer teaching critiques. The researcher also supported teachers in their classrooms on two occasions during the project. Using classroom observations of teaching, surveys about teachers' perspectives on ICTs, and interviews with teachers and students, Edwards-Groves documented writing instruction, use of technology, text construction, and teachers' and students' perspectives on the pedagogical changes made during the professional development project. Findings of the study demonstrated that the teachers' writing instruction changed over time; initially, teachers explicated a print-based, linear approach to writing, and computers were primarily used for word processing to publish students' work. Toward the end of the project, teachers and students were using ICTs to construct creative, multimodal texts geared toward authentic audiences. Over time, both teachers and students increased their knowledge and use of new literacies, which were motivating and fun to use. Students came to understand that writing in this digital environment required planning and design to create an effective and interesting text. Both teachers and students explicated their understandings of the value of interaction and collaboration in the teaching-learning process, which were facilitated by the use of ICTs. Collaborative writing experiences broke through cultural differences as students engaged in peer-editing and produced group projects where all participants had the opportunity for input. However, teachers expressed frustration over limited access to technology for instructional purposes. Nevertheless, results of the study illuminated the utility of ICTs in the writing program and highlighted the ways in which writing pedagogy in contemporary elementary classrooms is moving toward a multimodal approach to writing.

Mills and Exley (2014) elucidated how teachers confronted and worked through instructional tensions as they tried to assimilate and integrate new literacies practices from professional development training into their current writing programs. In the first year of a 2-year professional development program, the researchers introduced elementary school teachers to a multimedia writing program that focused on teaching students to design a range of multimodal and digitally-written texts (e.g., web pages, audiovisual podcasts, online comics, video documentaries, blogs). In the second year, the researchers used qualitative methods to document the teachers' implementation of the multimedia writing program with 4th grade students. Results indicated that the teachers' instruction initially reflected practices that are typical in print-based writing instruction. Teachers preferred direct instruction over student exploration. They strongly controlled the pacing of instruction and the products that students created. Instruction was oriented toward moving students quickly through the learning of new technology skills so students could create texts in new genres, as required by the writing curriculum. Initially, teachers believed the digital writing practices were too time-consuming and hindered their ability to meet the objectives of the mandated writing curriculum, but over time they consolidated new literacies instruction and writing instruction. Similarly, at first, teachers valued students' performance of digital literacies over quality of written products, but gradually they shifted to a more integrated, multimodal assessment of student writing. Despite the tensions these teachers faced, their new literacies instruction was motivating for students and highly successful in increasing time-on-task, in comparison with tasks that involved print-based writing without digital tools. As students appropriated the new literacies required to complete their writing assignments, they became active agents in the learning process; they also spent more time writing and engaged in more peer interactions and collaboration.

4.2.1.3. Access to ICTs. In six studies included in this review, researchers indicated that limited access to computers and ICTs also deterred teachers from using technology in the writing program. In some schools, computers were only available in computer labs, and often there was a shortage of computers in the lab (Ihmeideh, 2009). Teachers were less likely to provide opportunities for students to use computers in labs than in classrooms because of scheduling difficulties, the need to make advance reservations, and concerns over loss of instructional time as students moved to/from the classroom to the lab (Gremiger, 2003; Peterson & McClay, 2012). Portable laptops and tablets shared among teachers in a school were not always readily available when teachers needed them for instructional activities (Edwards-Grove, 2011). Even when computers or tablets were available in teachers' classrooms, large numbers of students and small numbers of computers/tablets created access challenges (Beam & Williams, 2015; Ihmeideh, 2009). Some teachers' classrooms were so small, or district finances so limited, they had only one computer in the classroom (Ihmeideh, 2009; Peterson & McClay, 2012). Lack of access to the Internet also was cited as a barrier to using technology for instructional activities (Ihmeideh, 2009). Teachers also voiced concerns regarding limited access to high-quality instructional software that was appropriate for their students' age/grade level (Ihmeideh, 2009; Vrasidas, 2015). A lack of institutional infrastructure and instructional technology (IT) support also was cited as a barrier to using technology for teaching and learning (Vrasidas, 2015). In one study, for example, the school district's IT staff failed to install the necessary software applications on student computers, despite the teacher's repeated requests for support (Beam & Williams, 2015).

4.2.1.4. RQ 3 synthesis: barriers to using technology. Teachers faced a number of challenges in implementing technology into the K-12 writing program. In some cases, their own beliefs about the negative impact of technology on the teaching-learning process made them reticent to integrate ICTs into writing instruction and assignments (Gremiger, 2003; Ihmeideh, 2009; Mills & Exley, 2014; Peterson & McClay, 2012; Vrasidas, 2015). In other cases, teachers were willing or motivated to use technology, but they lacked the necessary knowledge and skills, due to an absence of relevant professional development (Gremiger, 2003; Ihmeideh, 2009; Peterson & McClay, 2012; Sanchez, 2011; Vrasidas, 2015), or they had limited access to ICTs, high-quality instructional software, the Internet, and/or IT support (Beam & Williams, 2015; Edwards-Grove, 2011; Gremiger, 2003; Ihmeideh, 2009; Peterson & McClay, 2012; Vrasidas, 2015).

4.2.2. Additional findings

Our analysis revealed three additional findings related to the use of technology during writing instruction and related writing activities: The use of technology was motivating, encouraged collaboration, and was particularly supportive of reluctant writers. We discuss those findings in the sections to follow (see Table 1).

4.2.2.1. Technology and motivation. Thirteen studies included in this review highlighted the ways in which the use of ICTs to mediate the teaching and learning of writing was motivating for both students and teachers and led to increased student engagement and participation in instructional activities (e.g., Beam & Williams, 2015; Edwards-Groves, 2011; Gnach et al., 2007; Hitchcock et al., 2016; Howell et al., 2017; Jocius, 2013; Lawrence et al., 2009; Mills & Exley, 2014; Sessions et al., 2016; Voogt & McKenney, 2007; Yamac & Ulusoy, 2016; Yang et al., 2005). Students were motivated to participate in instructional activities because they wanted to use the technology, and their motivation-to-engage supported their learning.

In a recent study, Batsila and Tsihouridis (2016) *intentionally* used technology to motivate junior high school students to engage in reading and writing activities, with the ultimate goal of improving students' literacy skills and competence in English. A random sample of 51 students participated in the 8-week intervention in which they worked in small groups using Storyboard, a digital storytelling tool, to create digital stories about the topic of bullying. Using a pretest–posttest design, researchers documented statistically significant improvements in students' reading and writing skills at the end of the intervention. Results on a post-intervention questionnaire indicated that students found the digital storytelling tool easy to use. The tool fostered their interest and active participation in completing the assignment, and they were eager to use it again for future assignments. Students who once described writing as “boring” were now enthusiastic about writing activities. Interviews with teachers indicated that they believed the Storyboard tool was for easy for students to learn and use, created an active learner stance, encouraged collaboration and exchange of ideas among students, and supported students' literacy learning. Teachers also indicated that student collaboration using the digital tool was particularly supportive of struggling students' self-confidence. Both students and teachers described the technology as motivating.

4.2.2.2. Technology and collaboration. Eighteen of the studies we reviewed illuminated the ways in which the use of technology during writing instruction and related activities encouraged interaction and collaboration (Beam & Williams, 2015; Batsila & Tsihouridis, 2016; Edwards-Groves, 2011; Gnach et al., 2007; Hitchcock et al., 2016; Howell et al., 2017; Jocius, 2013; Lawrence et al., 2009; Mills & Exley, 2014; Sessions et al., 2016; Yamac & Ulusoy, 2016; Yang et al., 2005). As teachers and students learned to use a range of ICTs to mediate the teaching-learning process, they became co-teachers and co-learners (Edwards-Groves, 2011; Mills & Exley, 2014). Increased interaction and collaboration was a by-product of the use of technology, and it supported students' literacy learning. In fact, there was evidence to suggest that collaborative work was stronger in quality than individual work (Yamac & Ulusoy, 2016).

Three studies we reviewed investigated the use of ICTs to *intentionally* foster collaboration. In the earliest study, Chung and Walsh (2006) examined how young children's collaborative writing processes developed over time as they used computers to jointly compose stories. Kindergarten and first grade students worked in pairs to write three stories using the multimedia software program Kid Pix Studio Deluxe. Videotaped observations of the students collaboratively composing on the computer, interviews, and copies of

students' writing on the computer and in journals served as data sources. Analysis focused on the nature of peer interaction, including division of labor, and the learning opportunities that collaborative writing on computers afforded over time. Results demonstrated that the collaborative pairs gradually moved from dividing the work to a more integrative approach. For example, initially, children with greater literacy skills controlled the mouse and keyboard; over time, the pairs began to alternate control and coordinate their actions as they jointly composed their narratives. Results also demonstrated that the computer facilitated children's interactions and their collaborative writing by serving as object of reference. The computer helped to keep children on task and on track; for example, children referred to the computer screen to explain and explore ideas, sustain task-related discussions, and monitor their progress. The researchers argued that collaborative writing on the computer provided the children a relatively stress-free learning environment.

In a more recent study, [Pifarre and Fisher \(2011\)](#) examined collaboration in wikis to support elementary school students' composition strategies. Twelve 9- and 10-year-old students wrote collaboratively in the wiki environment to compose a common text about the feasibility of setting-up a colony on Mars. First, students participated in several lessons on how to engage in collaborative talk. Then, working in pairs, students conducted web-based research to gather information and write initial propositions. Next, students were taught how to write in the wiki environment, including how to add, delete, and revise text, and were introduced to a negotiation space where they could confer during the writing process, i.e., explain their ideas and reasons for changes they made in the common text. Over the next six sessions, student-pairs took turns working in the wiki to create the final collaborative text. Researchers analyzed each iteration of the group composition and found that students made both surface-level (e.g., spelling, punctuation) and text-based changes (e.g., additions, substitutions). They revised throughout the composing process as well as at the end, and their revisions demonstrated their concern for accuracy and meaning. Some revisions were grounded in students' knowledge of the topic; other changes were rooted in students' knowledge of the structure of texts. Collaboration in the wiki supported the development of composition and revision strategies by providing students a forum for sharing, discussing, and negotiating ideas, as well as opportunities to experience composing processes (i.e., planning, drafting, revising) and observe those same processes in their peers' efforts.

[Schillinger \(2011\)](#) examined the use of a wiki to mediate young girls' collaborative discussions of young adult novels. African American 8th-grade girls from an urban, low-income school and 7th-12th grade girls from a middle-class suburban school participated in the study. For four months, the young girls read two novels (each with a female protagonist), *The Skin I'm In* ([Flake, 1998](#)) and *Speak* ([Anderson, 1999\), completed weekly reading assignments, and responded to the novels in a reader response journal. To begin the online collaboration, their teachers posted prompts in the wiki space to facilitate introductions. Then, the students took over the discussion; they posted comparisons of the novels, comparisons of their lives vis-a-vis the novels, critiques of the novels, poetry and art they created inspired by the novels, and filmed scenes from the novels. Over 1000 responses were posted in the wiki space, revealing the ways in which the students collaboratively constructed knowledge about the novels and related topics. Schillinger analyzed those responses, and students' answers to pre- and post-surveys about their reading habits, growing up as a girl, and cultural factors, to investigate students' learning. Results revealed that the young girls frequently discussed bullying and the need to "stand up for yourself" and showed changes in the girls' perspectives about cultural differences between students from other schools. The novels provided students a shared experience, and the wiki provided an active platform for them to engage in critical reading, writing, and discussion with a responsive audience. Together, the literary tool and the technological tool mediated the students' collaborative construction of knowledge across cultural, geographical, and socio-economic boundaries.](#)

4.2.2.3. Technology and reluctant writers. Six of the studies we reviewed underscored the role of technology in supporting reluctant writers and students who struggle with literacy learning. As we noted earlier, for example, students who had been reluctant writers showed enthusiasm and wrote noticeably more when writing with digital tools ([Howell et al., 2017](#); [Yamac & Ulusoy, 2016](#)). Students who once described writing as difficult came to view it as less challenging when composing in a web-based writing environment ([Yang et al., 2005](#)). The composing process became less wily for struggling writers when they used the multimodal features of PowerPoint to organize their writing; this was particularly the case when students imported photos onto slides and used them as visual prompts for writing ([Hitchcock et al., 2016](#)). Similarly, the use of digital storyboards helped struggling writers understand the story-telling assignment and thus contribute to group work in ways that made them feel important, useful, and happy ([Batsila & Tsihouridis, 2016](#)). In their study of 5th-grade students' use of iPad applications for narrative writing, [Sessions et al. \(2016\)](#) found that several students who struggled with literacy learning demonstrated strong technology skills and could use the iPad apps in sophisticated ways; their status in the classroom was elevated and their self-esteem bolstered as they became peer-coaches for other students who needed technology support.

4.2.2.4. Synthesis of additional findings. The findings of our review demonstrated that students in kindergarten ([Beam & Williams, 2015](#); [Voogt & McKenney, 2007](#)), primary ([Yamac & Ulusoy, 2016](#)), elementary ([Edwards-Groves, 2011](#); [Gnach et al., 2007](#); [Mills & Exley, 2014](#); [Sessions et al., 2016](#); [Yang et al., 2005](#)), middle/junior high ([Batsila & Tsihouridis, 2016](#); [Hitchcock et al., 2016](#)), and high school ([Howell et al., 2017](#); [Jocius, 2013](#); [Lawrence et al., 2009](#)) were motivated to participate in writing instruction and related assignments because they wanted to use the technology. Moreover, the use of technology in the K-12 writing program promoted interaction and collaboration in ways that supported students' growth as readers and writers, as well as their ability to work with others ([Beam & Williams, 2015](#); [Batsila & Tsihouridis, 2016](#); [Chung & Walsh, 2006](#); [Edwards-Groves, 2011](#); [Gnach et al., 2007](#); [Hitchcock et al., 2016](#); [Howell et al., 2017](#); [Jocius, 2013](#); [Lawrence et al., 2009](#); [Mills & Exley, 2014](#); [Pifarre & Fisher, 2011](#); [Schillinger, 2011](#); [Sessions et al., 2016](#); [Yamac & Ulusoy, 2016](#); [Yang et al., 2005](#)). Technology-mediated instruction seemed particularly supportive of reluctant writers and students who struggled with literacy learning ([Batsila & Tsihouridis, 2016](#); [Hitchcock](#)

et al., 2016; Howell et al., 2017; Sessions et al., 2016; Yamac & Ulusoy, 2016; Yang et al., 2005).

5. Discussion and implications

5.1. Support for technology-mediated writing instruction

5.1.1. Students' growth as writers and learners

In light of the results of this review, we proffer a number of reasons for integrating technology into the K-12 writing program. First and foremost, the use of ICTs supported students' growth as writers and learners. Researchers reported improvements in kindergarten, primary, elementary, middle/junior, and high school students' writing (Doan & Bloomfield, 2014; Hitchcock et al., 2016; Holdich et al., 2004; Howell et al., 2017; Lawrence et al., 2009; Sessions et al., 2016), and some of these improvements were statistically significant (Chang et al., 2007; Doan & Bloomfield, 2014; McKenney & Voogt, 2009; Merrill & Rodriguez, 2005; Voogt & McKenney, 2007; Yamac & Ulusoy, 2016; Yang et al., 2005). The use of technology for writing assignments facilitated students' fluidity among the recursive phases of the writing process (e.g., Gnach et al., 2007; Kervin & Mantei, 2016; Turner & Katic, 2009).

Moreover, technology-mediated writing instruction supported the development of students' sequencing, analysis, synthesis, interpretative, problem solving, and generative thinking skills (Jocius, 2013; Kervin & Mantei, 2016; Mills, 2011; Sessions et al., 2016; Yamac & Ulusoy, 2016)—skills that are essential to students' continued learning. Improvements in students' new literacies skills also were reported (Edwards-Groves, 2011; Merrill & Rodriguez, 2005; Mills & Exley, 2014; Yamac & Ulusoy, 2016).

5.1.2. Motivation

Second, the use of technology in the K-12 writing program was motivating. From kindergarten to high school, students were motivated to participate in writing activities because they wanted to use the technology. They were attentive during instruction and actively engaged while completing writing assignments (e.g., Beam & Williams, 2015; Hitchcock et al., 2016; Howell et al., 2017; Lawrence et al., 2009; Mills & Exley, 2014; Sessions et al., 2016). Some students explicitly stated that they preferred technology-mediated writing activities over print-based instruction and assignments (e.g., Howell et al., 2017; Jocius, 2013; Peterson & McClay, 2012). Students who are motivated to participate in instructional activities are typically on task, cognitively engaged, and responding affectively to the activity; their motivation drives their learning (Malloy, Marinak, Gambrell, & Mazzoni, 2013; McGeown, Osborne, Warhurst, Norgate, & Duncan, 2016).

5.1.3. Social interaction and collaboration

Third, the use of technology encouraged social interaction and collaboration around text construction. Collaborative composing required students to organize how they would work together and share responsibility for specific tasks—skills that are valued both in school and in the workplace (Chung & Walsh, 2006; Sessions et al., 2016). Conferring with each other as they worked generated important conversations about composing processes and products, enhancing students' collaborative knowledge construction and their learning experience (e.g., Batsila & Tsihouridis, 2016; Edwards-Groves, 2011; Pifarre & Fisher, 2011). Teachers reported that students benefitted from the increased social interaction and collaboration (e.g., Edwards-Groves, 2011; Hitchcock et al., 2016). In some cases, collaborative work was stronger in quality than individual work (Yamac & Ulusoy, 2016). Social interaction played a significant role in students' multimodal and digital writing experiences, just as it has in print-based writing (e.g., Agee & Altarriba, 2009; Dyson, 2003; Williams, 2018; Williams & Hufnagel, 2005). Importantly, working collaboratively with peers helped students to break through cultural differences (Edwards-Groves, 2011; Schillinger, 2011).

5.1.4. Reluctant writers

Our fourth and final reason is that the use of technology for writing was especially supportive of reluctant and struggling writers. Reluctant writers demonstrated enthusiasm for participating in digital writing. They no longer found the process “boring” because they were writing with “fun” digital tools and collaborating with peers (e.g., Batsila & Tsihouridis, 2016; Howell et al., 2017; Yamac & Ulusoy, 2016). Technology inherently has a variety of assistive supports and scaffolds that help struggling writers to persist when the composing process proves challenging (Hitchcock et al., 2016; Peterson & McClay, 2012; Yang et al., 2005). Those scaffolds made it possible for struggling writers to interact and collaborate with peers in ways that increased their self-confidence (Batsila & Tsihouridis, 2016; Yamac & Ulusoy, 2016). This was particularly the case when struggling writers were technology savvy (Sessions et al., 2016).

5.2. Implications for pedagogy and research

5.2.1. A process approach to writing

The results of this review offer strong support for a process approach to writing instruction. In 14 of the studies included in this review, researchers grounded their investigations in the process approach framework (Gnach et al., 2007; Pifarre & Fisher, 2011; Sanchez, 2011; Turner & Katic, 2009; Yang et al., 2005), or teachers used a process approach to teach writing (Beam & Williams, 2015; Chang et al., 2007; Doan & Bloomfield, 2014; Edwards-Groves, 2011; Kervin & Mantei, 2016; Hitchcock et al., 2016; Howell et al., 2017; Peterson & McClay, 2012; Yamac & Ulusoy, 2016). Within the supportive infrastructure of the process approach to writing, teachers and researchers introduced and integrated a range of multimedia and digital tools into the K-12 writing program. Teachers provided new literacies instruction, encouraged student agency, and scaffolded students' experimentation with multimodal,

digital writing. As students designed, produced, and presented multimodal texts to represent their learning, their teachers reconceptualized the writing process and writing instruction to account for these multimodal composing processes. Teachers and students successfully blended print-based literacies and new literacies practices and products (Leander, 2009). Multimedia and multimodality did not replace print-based literacies or conventional writing skills but rather extended the writing process to account for the new literacies that technology affords (Beam & Williams, 2015; Edwards-Groves, 2011). These 14 studies demonstrate that a process approach aligns with a multiliteracies framework by conceptualizing writing as a recursive process of multimodal design and composition (New London Group, 1996).

Furthermore, the process approach to writing is a well-researched, established, and widely-used framework for writing instruction (Applebee & Langer, 2013; Graham & Sandmel, 2011; Pritchard & Honeycut, 2006). The instructional approach has changed over time, moving from a linear, step-by-step model to a recursive model of the writing process—i.e., movement across and among processes of planning, drafting, receiving feedback, revising, and editing—all in route to sharing or publication. Students engage regularly in extended writing for authentic purposes and receive support through peer interaction, explicit instruction on strategies that support recursive writing, mini-lessons, and conferencing (Applebee & Langer, 2013; Pritchard & Honeycut, 2006). Considerable research has documented the usefulness of the process approach for improving the quality of students' writing (Graham & Perin, 2007; Graham & Sandmel, 2011). Learning to write and developing competence as a writer requires fundamental understandings of writing processes and the acquisition of a range of writing strategies and skills, including the use of technology. We would argue that a process approach to writing, as defined here, is an appropriate framework within which to integrate ICTs and develop instructional activities designed to advance students' 21st-century writing skills.

5.2.2. Teacher professional development and institutional support

The results of this review illuminate the need for high-quality teacher professional development on the use of ICTs in the writing program. In seven studies included in this review, researchers revealed or teachers expressed a need for professional development on how to meaningfully integrate technology into the writing curriculum (Edwards-Groves, 2011; Gremiger, 2003; Ihmeideh, 2009; Mills & Exley, 2014; Peterson & McClay, 2012; Sanchez, 2011; Vrasida, 2015). To be authentic, professional development must reflect teachers' immediate contexts and everyday instructional practices (Cochran-Smith & Lytle, 2009). Teachers need opportunities to develop their knowledge of multimodality, including linguistic, visual, audio, and spatial elements, to more fully comprehend digital design and composition using a range of media (Dalton, Proctor, Uccelli, Mo, & Snow, 2011). They need conversations with other professionals that speak to the tensions they feel between addressing curricular objectives and integrating technology into the writing program. Teachers need occasions to examine, and perhaps disentangle, some of their assumptions about the nature of the writing process and composing with computers and digital tools. In addition, teachers must be afforded sufficient time and institutional support to plan and develop technology-mediated instruction and writing activities that exploit the affordances of ICTs. Institutional support also will be needed to ensure the availability of computers and access to ICTs in every classroom; otherwise, the impact of technology on instruction and student outcomes will likely be marginal (Beam & Williams, 2015; Becker, 2000; Zheng, Warschauer, Lin, & Chang, 2016). Supporting the professional development of teachers will go a long way toward instantiating pedagogical practices that engage students in using technology to compose dynamic multimodal and digital texts that represent their learning and communicate their ideas, perspectives, and intentions.

5.2.3. Directions for future research

Meaningful integration of technology into the K-12 writing program also will require further research on the ways teachers can use multimedia and digital tools to teach writing in this digital age, meet the objectives of the writing curriculum, and engage all students in the composing process, including reluctant and struggling writers. Substantial investigative work is needed in the early childhood program; it has received limited research attention thus far. Most children begin the learning-to-write process at home by imitating the writing practices of their parents and family members, but it is in school settings that children come to understand the nature of the composing process and solidify their writing skills (Mackenzie & Petriwskyj, 2017). In particular, future research needs to explore young children's experimentation with digital writing and their text production processes (Merchant, 2007). Continued exploration of how digital writing environments and online writing communities can support students' competence as writers also is needed.

Conflicts of interest

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